CRF State Summary v10

K25 + Two-Layer Pinning Session — April 2026

Maturity: 8.0/8 Theory · 7.5/8 Empirical gate open

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1 Primitive Chain

$\delta \to P \to C \to D_{\mathrm{KL}} \to \theta \to R \to t$

With $n = 2^{d_s}$: master chain $\delta \to n \to D_0 = n(1 - e^{-1/n}) \to \beta = \ln n / D_0,\ \kappa = \sqrt{d_s}\alpha/n$.

2 New This Session

ID	Identity / Finding	Gap	Significance
K25	$K^* \varphi_{\mathrm{var}}/(F\sigma_\Psi) = e/\sqrt{\varphi}$	0.004%	Corollary K20÷K21
K21 struct	ε_0 cancels in K21	exact	K21 is pure $f(n)$
Pinning L1	$f(8) = \sqrt{\varphi}$ pins $n = 8$	0.004%	Golden ratio selects n
Pinning L2	K20 derives $\varepsilon_0 = 0.007550$	0.005%	Euler's e fixes ε_0
Structure	$(\varphi, e) \to (n, \varepsilon_0) \to$ all	—	Two primitives from two constants

3 The Master System

Layer 1 (K21, ε_0-free): $f(n) = 4n^3(1-e^{-1/n})(n+2)/[(n-1)\sqrt{2n+3}(n^2-2)] = \sqrt{\varphi}$ $\Rightarrow n = 8$ (unique integer solution)

Layer 2 (K20, $n = 8$ + Euler e): $K^* T_{\mathrm{avg}}/F = e \ \Rightarrow\ \varepsilon_0 = 0.007550$

$(\varphi, e) \to (n, \varepsilon_0) \to$ **everything:** $D_0, K^*, \beta, \kappa, \sigma_\Psi, \psi_{\mathrm{grad}}, F, H_a, H_b, T_{\mathrm{avg}}, \ldots$

4 Complete Status Table

Result	Status	Gap	Note
Foundations			
$n = 3$ from SO(3)	Derived	—	#16
$\alpha = \ln 2$	Derived	—	
Schwarzschild, $G = \ln 2 \cdot c^2/D_0$	Derived	—	Paper v4
$\ln(n)/D_0$ *lattice*			
$D_0 = n(1 - e^{-1/n})$ (K2)	Derived	0.03%	
$K^* = D_0/n$ (K1)	Derived	0.43%	
$\beta D_0 = \ln n$ (K11)	Derived	0.005%	
$\kappa = \sqrt{d_s} \ln 2/n$ (K7)	Hypothesis	0.047%	
$\kappa/K^* = \beta/\sqrt{d_s}$ (K17)	Derived	exact	
$\beta/\kappa = n\sqrt{d_s}/D_0$ (K18)	Derived	exact	
$\beta\kappa = \ln^2 n/(n\sqrt{d_s}D_0)$ (K19)	Derived	0.005%	
Pinning — n and ε_0			
$n = 2^{d_s}$ (binary Born rule)	Physical arg	exact	
K21 ε_0-independent	**Proved**	exact	New: ε_0 cancels
K21 pins $n = 8$: $f(8) = \sqrt{\varphi}$	Hypothesis	0.004%	New
K20 derives ε_0 from e	Hypothesis	0.005%	New
$(\varphi, e) \to (n, \varepsilon_0)$	Structural	0.005%	Two-layer
e-web			
$K^* T_{\text{avg}}/F = e$ (K20)	Hypothesis	0.005%	
$H_u\beta/H_b = e$ (K22)	Hypothesis	0.041%	
φ-web			
$T_{\text{avg}}\sigma_\Psi/\varphi_{\text{var}} = \sqrt{\varphi}$ (K21)	Hypothesis	0.004%	
$m = \varphi$ (K6)	Hypothesis	0.06%	
$F\varphi^2 = 4$ (K14)	Hypothesis	0.057%	
$\sqrt{6}$-web			
$H_a\beta/\varphi = \sqrt{6}$ (K23)	Hypothesis	0.044%	
$n\psi_{\text{grad}}/K^* = \sqrt{6}$ (K24)	Hypothesis	0.046%	

5 Key Numbers v10

$\alpha = \ln 2$	$\varepsilon_0 = 0.00755$	$D_0 = 0.94003$	$\kappa = 0.15007$	$K^* = 0.11750$
$n = 8 = 2^3$	$N = 500$	$d_s = 3.031$	$\beta = 2.2121$	$F = 1.5378$
$T_{\text{avg}} = 35.57$	$\varphi = 1.61803$	$p = 9/10$	$H_a = 1.7925$	$H_b = 1.6717$
$\varphi_{\text{var}} = 0.4844$	$\sigma_\Psi = 0.01732$	$\psi_{\text{grad}} = 0.03600$	$H_u = 2.055$	$e = 2.71828$

6 Open Gaps (Priority Order)

Priority 1: $m(m-1) = 1$ energy balance

$m(m-1) = \beta\kappa/f(\delta)$ has $m = \varphi$ when $f = \beta\kappa$. Show $f = \beta\kappa$ from steady-state dynamics without fixing $s_{\max}$.

Priority 2: K21 exact ($f(8) = \sqrt{\varphi}$ algebraically)

The equation $4 \times 8^3(1 - e^{-1/8}) \times 10/(7\sqrt{19} \times 62) = \sqrt{\varphi}$ has 0.004% gap. Prove or disprove it as exact.

Priority 3: K7 physical proof ($\kappa = \sqrt{d_s} \ln 2/n$)

Derive from Born resampling that the mass transfer per event is $\sqrt{d_s}\alpha/n$.

Priority 4: P0-H Subject 4 (empirical gate)

Flat 50% accuracy across all RT bins for blocked-type subject.

7 Documents This Session

CRF_KStar_Closure.pdf	K^* gap; fcorr derived; D0=1-nε_0
CRF_K11K12K14.pdf	K11/K12/K14; $n = 2^{d_s}$ structural key
CRF_MasterSystem_K17K19.pdf	K17/K18/K19; p derived; $\ln(n)/D_0$ lattice
CRF_StateSummary_v8.pdf	State after K11–K19
CRF_K25_Structure.pdf	K25; K21 ε_0-free; two-layer pinning
CRF_StateSummary_v10.pdf	This document

K21 asks: what n gives $\sqrt{\varphi}$? The answer is 8.
K20 asks: what ε_0 gives e? The answer is 0.007550.
φ selects the symmetry. e calibrates the noise.
Together they build the world.